

AI Methodology Appendix

P.R.I.M.E. Documentation of AI Interactions — Zehan Qin | ECON 5200 | Spring 2026

This appendix documents significant AI interactions during the development of this consulting report, following the **P.R.I.M.E. framework**: Prompt, Response, Iterate, Modify, Evaluate. The primary AI tool used was Claude (Anthropic); a secondary AI copilot surfaced one methodological issue that was subsequently cross-validated with Claude and the original literature. All substantive decisions — method selection, result interpretation, threat analysis, and policy recommendations — were made by the human analyst after verifying AI outputs against authoritative sources.

Entry 1 — Code Generation: Data Loading & EDA Pipeline

Prompt: "Help me load the Lalonde dataset by combining NSW treated with CPS comparison group, and generate EDA visualizations including balance checks and distribution plots."

Response: Claude generated Python code using the `causaldata` package to load `nsw_mixture` and `cps_mixture`, combined them into an observational dataset (N=16,177), and produced balance tables with standardized mean differences plus three visualization plots (outcome distribution, pre-treatment earnings, covariate imbalance).

Iterate: Refined the color scheme and axis labels for clarity. Asked for standardized mean difference calculation using the proper pooled within-group standard deviation formula rather than the overall sample SD, which is a non-standard shortcut.

Modify: Added interpretation text below each visualization. Adjusted bin counts for histograms to better show the distributional differences. Switched the SMD formula from overall SD to the standard pooled within-group SD.

Evaluate: Verified dataset dimensions match published Lalonde literature (185 treated, 15,992 CPS controls). Cross-checked summary statistics against Dehejia & Wahba (1999). The SMD values using the corrected formula (e.g., `black` = 2.43, `re74` = -1.57) align with standard reports in the Lalonde replication literature.

Entry 2 — Analysis Assistance: DML Implementation

Prompt: "Implement Double Machine Learning using `econml`'s `LinearDML` to estimate the ATE of job training on earnings, using GBR nuisance models."

Response: Claude generated DML code with `GradientBoostingRegressor` for both outcome and treatment models, 5-fold cross-validation, and ATE with confidence intervals.

Iterate: Asked for a robustness check with Random Forest nuisance models. Also requested a comparison plot of naive vs. causal estimates to visually demonstrate the bias correction.

Modify: Changed `max_depth` from default to 4 to reduce overfitting in nuisance models. Added the experimental RCT benchmark (\$1,794) to the comparison plot as a visual validation target.

Evaluate: Compared the DML ATE to published results in Lalonde replication literature. Confirmed the estimate moves in the correct direction (positive) relative to the naive estimate (negative). The experimental benchmark serves as an external validation check — a rare advantage of this dataset that enables direct validation of observational methods.

Entry 3 — Writing: Threats to Identification

Prompt: "Help me draft the Threats to Identification section, focusing on unobserved confounding, overlap violations, and sensitivity to comparison group choice."

Response: Claude outlined three threats with bias direction analysis and proposed mitigations.

Iterate: Asked for more specificity on the overlap problem and what trimming approaches could address it. Requested addition of adversarial angles (specification sensitivity, external validity, ROI) to meet the rubric requirement of addressing adversarial questions.

Modify: Rewrote the bias direction analysis for the unobserved confounding threat to acknowledge ambiguity rather than claim a definitive direction. Added reference to the Lalonde literature showing sensitivity to comparison group construction. Added a dedicated adversarial angles subsection covering specification, external validity, and ROI.

Evaluate: Cross-checked against Smith & Todd (2005) discussion of comparison group sensitivity. Verified that the threats are consistent with standard critiques in the observational causal inference literature. All bias direction claims are logically consistent with the observed data and standard econometric reasoning.

Entry 4 — Methodology Refinement: DML Treatment Specification

Prompt: "Another AI copilot flagged that my DML code uses a Regressor for `model_t`, but my treatment is binary (0/1). It suggested I should use a Classifier for `model_t` and add `discrete_treatment=True`. Is this critical, and how do I fix it properly?"

Response: Claude confirmed this is a methodology-level issue rather than a coding style preference. Because `treat` is binary, `model_t` is estimating the propensity score $P(T=1|X)$, which must output probabilities bounded in $[0,1]$. Using a Regressor can produce predictions outside this range and violates the theoretical convergence guarantees of DML (Chernozhukov et al., 2018). Claude provided the corrected code: replacing `GradientBoostingRegressor` and `RandomForestRegressor` in `model_t` with their Classifier counterparts, and adding `discrete_treatment=True` to the `LinearDML` call.

Iterate: Asked Claude to explain why this matters more for this specific dataset. The response noted that Lalonde has a severely imbalanced treatment assignment (only ~1.1% treated), which makes the propensity score estimation especially sensitive to model specification.

Modify: Applied the fix to both the main DML estimation and the robustness check. Added `from sklearn.ensemble import GradientBoostingClassifier, RandomForestClassifier` to the imports. Re-ran all downstream cells (comparison plot, Executive Summary numbers) to reflect the corrected estimates.

Evaluate: Cross-validated the fix by consulting Claude independently rather than accepting the first copilot's suggestion at face value. Verified the corrected estimates are closer to the experimental RCT benchmark of \$1,794, which is the expected behavior of a correctly-specified DML with a binary treatment. Confirmed via the `econml` documentation that `discrete_treatment=True` is the recommended setting for binary treatments. Cross-checked against Chernozhukov et al. (2018) Theorem 3.1, which requires the propensity score estimator to be consistent in the sup-norm — a property Classifiers satisfy but Regressors do not for binary outcomes. The fact that the corrected RF estimate (\$1,541) is much closer to the experimental benchmark (\$1,794) than the original estimate provides empirical validation that the methodological correction was justified.

Entry 5 — Expression Accuracy and Consistency Audit

Prompt: "A reviewer flagged several expression-accuracy issues across my notebook and PDF deliverables: (1) the Visualization 3 interpretation says *All covariates show substantial imbalance* but

my data shows hisp has $|SMD| = 0.05$, below the threshold; (2) several places use *ground truth*, *near-zero*, and *exact opposite of the truth* which are either overclaims or fact-misstatements; (3) the comparison plot gives the RCT benchmark a hand-coded $\pm\$800$ error bar that is not a real confidence interval. Should I change all of these?"

Response: Claude evaluated each issue separately rather than blanket-accepting them. Classified them as: (1) **factual error** — *All covariates* contradicts the actual SMD table, must fix; (2) **style/rigor issue** — *ground truth* is actually common in the causal inference literature, but for an academic report *experimental benchmark* is more precise; *near-zero* is a factual overstatement since \$2,096 is low but not near zero; (3) **presentation error** — the $\pm\$800$ band implies a confidence interval that was never estimated, so it should be replaced with a plain reference line.

Iterate: Asked Claude to scan all project files (notebook, five PDFs, Streamlit app.py) systematically for any additional instances of the same expression patterns, to avoid missing occurrences during manual review.

Modify: Applied changes across all files: (a) notebook Visualization 3 text changed from *All covariates* to *Most covariates*; (b) all instances of *ground truth* replaced with *experimental benchmark* or *RCT benchmark* in notebook, Technical Report, AI Appendix, Research Proposal, and Streamlit app; (c) *near-zero pre-treatment earnings* replaced with *very low pre-treatment earnings*; (d) the comparison plot was restructured to show the RCT benchmark as a reference line rather than as a bar with fake error bars; (e) the Technical Report caption *every covariate in our sample exceeds this* was corrected to *most covariates exceed this threshold (the exception is hisp, with $|SMD| = 0.05$)*.

Evaluate: Used programmatic verification rather than manual review: ran a text search across all output files for each problematic phrase and confirmed zero occurrences in all files after the fix. This automated consistency check is more reliable than human scanning across multiple documents. The final deliverables now use uniform, precise terminology: *experimental benchmark* for the RCT reference value, *very low* (not *near-zero*) for the treated group's pre-treatment earnings, and *most covariates* (not *all*) when describing balance-check results. The episode reinforced a methodology lesson: AI feedback should be filtered through the analyst's judgment — not every suggestion is equally important, and distinguishing factual errors from style preferences is itself a professional skill.

Human judgment exercised throughout. AI tools accelerated code generation, surfaced a methodology bug, and assisted with drafting. All final decisions — including the choice of DML over alternative methods, the interpretation of two differing DML specifications, the honest acknowledgment of limitations, and the policy recommendation — were made by the human analyst after verifying AI outputs against authoritative sources (Chernozhukov et al. 2018; Smith & Todd 2005; Dehejia & Wahba 1999; econml documentation).